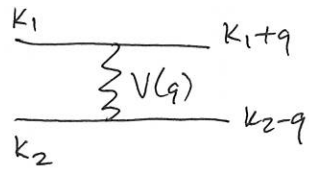
RPA revisited

$$H_{int} = \frac{1}{2V} \sum_{\substack{k_1 k_2 q \\ \sigma_1 \sigma_2}} V(\bar{q})\, c^+_{k_1+q\,\sigma_1}\, c^+_{k_2-q\,\sigma_2}\, c_{k_2\sigma_2}\, c_{k_1\sigma_1}$$



Coulomb interaction in $D=3$

$$V(\bar{r}) = \frac{e^2}{r} \Rightarrow V(\bar{q}) = \frac{4\pi e^2}{q^2}$$

However, for neutral atoms (He^3, cold atoms in traps) the interaction is short-ranged. Can be often approximated by contact potential $V(\bar{r}) = U\,\delta(\bar{r})$.

So in real space

$$\int dr_1 dr_2 \underbrace{V(\bar{r}_1-\bar{r}_2)}_{U\,\delta(\bar{r}_1-\bar{r}_2)} c^+_{r_1\sigma_1}\, c^+_{r_2\sigma_2}\, c_{r_2\sigma_2}\, c_{r_1\sigma_1} =$$

$$= U \int dr\, c^+_{r\sigma_1}\, c^+_{r\sigma_2}\, c_{r\sigma_2}\, c_{r\sigma_1} = 0 \text{ for } \sigma_1 = \sigma_2 \text{ (Pauli principle)}$$

$\Rightarrow$ for contact interaction only $\sigma_2 = -\sigma_1$ is present.

$\Rightarrow$ let us write $V(\bar{r}) = V_{\parallel}(\bar{r})\,\delta_{\sigma_1\sigma_2} + V_{\perp}(\bar{r})\,\delta_{\sigma_1,-\sigma_2}$

$$H_{int} = \frac{1}{2}\sum_{\sigma_1\sigma_2}\int dr_1 dr_2 \Big[V_{\parallel}(r_1-r_2)\, c^+_{r_1\sigma_1}\, c^+_{r_2\sigma_2}\, c_{r_2\sigma_1}\, c_{r_1\sigma_1} + V_{\perp}(r_1-r_2)\, c^+_{r_1\sigma_1}\, c^+_{r_2-\sigma_1}\, c_{r_2-\sigma_1}\, c_{r_1\sigma_1} \Big]$$

~~Short Coulomb interaction: $V_{\parallel}$~~

Observe that if $V_{\perp}(\bar{r}) = U\,\delta(\bar{r})$, then

$$\sum_{\sigma} \frac{1}{2} U \int d\bar{r}\, c^+_{r\sigma}\, c^+_{r-\sigma}\, c_{r-\sigma}\, c_{r\sigma} = \sum_{\sigma} \frac{1}{2} U \int dr\, c^+_{r\sigma} c_{r\sigma}\, c^+_{r-\sigma} c_{r-\sigma}$$

$$= \frac{1}{2} U \int dr\, \hat{n}_{r\uparrow}\hat{n}_{r\downarrow} + \hat{n}_{r\downarrow}\hat{n}_{r\uparrow} = U \int d\bar{r}\, \hat{n}_{r\uparrow}\hat{n}_{r\downarrow}$$

this is Hubbard interaction.

In momentum space

$$H_{int} = \frac{1}{2V} \sum_{\sigma} \sum_{k_1 k_2 q} \Big(V_{\parallel}(\bar{q})\, c^{+}_{k_1+q\,\sigma}\, c^{+}_{k_2-q\,\sigma}\, c_{k_2\sigma}\, c_{k_1\sigma}$$

$$+ V_{\perp}(\bar{q})\, c^{+}_{k_1+q\,\sigma}\, c^{+}_{k_2-q,-\sigma}\, c_{k_2,-\sigma}\, c_{k_1\sigma} \Big)$$

Coulomb : $V_{\parallel}(\bar{q}) = V_{\perp}(\bar{q})$

Contact : $V_{\parallel}(\bar{q}) = 0.$

RPA as mean-field approximation:

Start with spinless ~~electron~~ particles, $H = H_0 + H_{int}$

$$H_{int} = \frac{1}{2}\int dr_1 dr_2 \; V(\bar{r}_1 - \bar{r}_2)\; \bar{\rho}(r_1)\,\bar{\rho}(r_2)$$

where $\bar{\rho}(r) = \rho(r) - \rho_0$

operators — average charge density

Assume that uniform state is stable $\Rightarrow$ in the absence of any additional perturbations the ground state of $\hat{H}$ is such that $\langle \bar{\rho}(r) \rangle = 0$.

We now add external time-dependent perturbation

$$H'(t) = \int d\bar{r}\; \varphi(\bar{r}, t)\; \bar{\rho}(r)$$

We now expect that $\langle \bar{\rho}(r) \rangle_t \neq 0$. This notation means expectation value of operator $\bar{\rho}(r)$ with respect to the full Hamiltonian $\hat{H} + \hat{H}'(t)$ $\Rightarrow$ hence the expectation value $\langle \bar{\rho}(r) \rangle_t$ should be t-dependent.

To separate this induced by $H'(t)$ average, write $\bar{\rho}(r) = \langle \bar{\rho}(r) \rangle_t + \delta\bar{\rho}(r)$ and <u>approximate</u> $\hat{H}_{int}$ by linear in fluctuations $\delta\bar{\rho}(r)$ form:

$$H_{int} = \frac{1}{2}\int dr_1 dr_2 \; V(r_1 - r_2)\left[\langle \bar{\rho}(r_1)\rangle_t + \delta\bar{\rho}(r_1)\right]\left[\langle \bar{\rho}(r_2)\rangle_t + \delta\bar{\rho}(r_2)\right]$$

$$= \frac{1}{2}\int dr_1 dr_2 \; V(r_1 - r_2)\Big\{ \langle \bar{\rho}(r_1)\rangle_t \langle \bar{\rho}(r_2)\rangle_t +$$

$$+ \underbrace{\delta\bar{\rho}(r_1)}_{\bar{\rho}(r_1) - \langle \bar{\rho}(r_1)\rangle_t} \langle \bar{\rho}(r_2)\rangle_t + \underbrace{\delta\bar{\rho}(r_2)}_{\bar{\rho}(r_2) - \langle \bar{\rho}(r_2)\rangle_t} \langle \bar{\rho}(r_1)\rangle_t + \cancel{\delta\bar{\rho}(r_1)\cdot\delta\bar{\rho}(r_2)} \Big\}$$

assume this term small

$$\Rightarrow H_{int} \approx \frac{1}{2}\int dr_1 dr_2\, V(r_1-r_2)\Big\{ \bar{\rho}(r_1)\langle\bar{\rho}(r_2)\rangle_t + \bar{\rho}(r_2)\langle\bar{\rho}(r_1)\rangle_t -$$

$$\underbrace{-\langle\bar{\rho}(r_1)\rangle_t\ \langle\bar{\rho}(r_2)\rangle_t}_{\text{not an operator}}\Big\} = \int d\vec{r}_1\ \bar{\rho}(r_1)\int d\vec{r}_2\ V(\vec{r}_1-\vec{r}_2)\langle\bar{\rho}(r_2)\rangle_t .$$

$+$ constant.

In this mean-field (MF) approximation

$$H + H'(t) = H_0 + \int d\vec{r}\ \bar{\rho}(r)\Big(\varphi(\vec{r},t) + \underbrace{\int d\vec{r}_2\ V(\vec{r}-\vec{r}_2)\langle\bar{\rho}(r_2)\rangle_t}_{\gamma(\vec{r},t)}\Big)$$

So that time-dependent perturbation is produced by the combined potential $\underbrace{\varphi(\vec{r},t)}_{\text{external}} + \underbrace{\gamma(\vec{r},t)}_{\text{due to interaction between particles of the system}}$

By <u>linear response</u>

$$\langle\bar{\rho}(\vec{r})\rangle_t = \int d\vec{r}'\, dt'\ \chi_0(\vec{r}-\vec{r}', t-t')\left[\varphi(\vec{r}',t) + \gamma(\vec{r}',t')\right]$$

with $\chi_0(\vec{r},t) = -i\,\theta(t)\,\langle[\bar{\rho}(\vec{r},t), \bar{\rho}(0,0)]\rangle_{H_0}$

F.T. gives

$$\langle\bar{\rho}(\vec{q},\omega)\rangle = \chi_0(\vec{q},\omega)\Big(\varphi(\vec{q},\omega) + \gamma(\vec{q},\omega)\Big)$$

But $\gamma(\vec{q},\omega) = V(\vec{q})\langle\bar{\rho}(\vec{q},\omega)\rangle$

$$\langle\bar{\rho}(\vec{q},\omega)\rangle = \chi_0(\vec{q},\omega)\ \varphi(\vec{q},\omega) + \chi_0(\vec{q},\omega)\ V(\vec{q})\ \langle\bar{\rho}(\vec{q},\omega)\rangle$$

$$\langle\bar{\rho}(\vec{q},\omega)\rangle = \frac{\chi_0(\vec{q},\omega)}{1-V(\vec{q})\chi_0(\vec{q},\omega)}\ \varphi(\vec{q},\omega) \equiv \chi(\vec{q},\omega)\ \varphi(\vec{q},\omega)$$

Charge susceptibility of interacting system χ

$$\chi(\vec{q},\omega) = \frac{\chi_0(\vec{q},\omega)}{1 - V(\vec{q})\,\chi_0(\vec{q},\omega)} \qquad \underline{\underline{RPA}}$$

is expressed in terms that of non-interacting χ_0.

This is our previous result: ~O~ + ~O~O~ + ~O~O~O + ...

Next Extention to <u>spinful particles</u>:

let $H'(t) = \int d\vec{r}\; \varphi_\uparrow(\vec{r},t)\,\bar{\rho}_\uparrow(\vec{r}) + \varphi_\downarrow(\vec{r},t)\,\bar{\rho}_\downarrow(\vec{r})$

Hence MF approximation now gives

$$H + H'(t) \to H_0 + \int d\vec{r}\Big\{ \bar{\rho}_\uparrow(\vec{r})\big(\varphi_\uparrow(\vec{r},t) + \gamma_\uparrow(\vec{r},t)\big) + \bar{\rho}_\downarrow(\vec{r})\big(\varphi_\downarrow(\vec{r},t) + \gamma_\downarrow(\vec{r},t)\big)\Big\}$$

where $\gamma_\uparrow(\vec{r},t) = \int d\vec{r}_2\; V_{\parallel}(\vec{r}-\vec{r}_2)\langle\bar{\rho}_\uparrow(\vec{r}_2)\rangle_t + V_\perp(\vec{r}-\vec{r}_2)\langle\bar{\rho}_\downarrow(\vec{r}_2)\rangle_t$

$\gamma_\downarrow(\vec{r},t) = \int d\vec{r}_2\; V_{\parallel}(\vec{r}-\vec{r}_2)\langle\bar{\rho}_\downarrow(\vec{r}_2)\rangle_t + V_\perp(\vec{r}-\vec{r}_2)\langle\bar{\rho}_\uparrow(\vec{r}_2)\rangle_t$

F.T: $\begin{cases} \gamma_\uparrow(\vec{q},\omega) = V_{\parallel}(\vec{q})\langle\bar{\rho}_\uparrow(q,\omega)\rangle + V_\perp(\vec{q})\langle\bar{\rho}_\downarrow(\vec{q},\omega)\rangle \\ \gamma_\downarrow(\vec{q},\omega) = V_{\parallel}(\vec{q})\langle\bar{\rho}_\downarrow(\vec{q},\omega)\rangle + V_\perp(\vec{q})\langle\bar{\rho}_\uparrow(\vec{q},\omega)\rangle \end{cases}$

Linear response

$$\langle\bar{\rho}_\uparrow(\vec{r})\rangle_t = \int d\vec{r}'\,dt'\; \chi_{0\uparrow}(\vec{r}-\vec{r}',t-t')\big(\varphi_\uparrow(\vec{r}',t') + \gamma_\uparrow(\vec{r}',t')\big)$$

$\hookrightarrow -i\,\theta(t-t')\,\langle[\bar{\rho}_\uparrow(r,t),\bar{\rho}_\uparrow(r',t')]\rangle_{H_0}$

$$\langle\bar{\rho}_\downarrow(\vec{r})\rangle_t = \int d\vec{r}'\,dt'\; \chi_{0\downarrow}(\vec{r}-\vec{r}',t-t')\big(\varphi_\downarrow(\vec{r}',t') + \gamma_\downarrow(\vec{r}',t')\big)$$

$\hookrightarrow -i\,\theta(t-t')\,\langle[\bar{\rho}(\vec{r},t),\bar{\rho}(\vec{r}',t')]\rangle_{H_0}$

$$\langle \bar{\rho}_\uparrow(q,\omega)\rangle = \chi_{0\uparrow}(\bar{q},\omega)\Big(\varphi_\uparrow(q,\omega) + V_{\parallel}(\bar{q})\,\langle \bar{\rho}_\uparrow(q,\omega)\rangle + V_{\perp}(\bar{q})\langle \bar{\rho}_\downarrow(q,\omega)\rangle\Big)$$

$$\langle \bar{\rho}_\downarrow(q,\omega)\rangle = \chi_{0\downarrow}(\bar{q},\omega)\Big(\varphi_\downarrow(q,\omega) + V_{\parallel}(\bar{q})\,\langle \bar{\rho}_\downarrow(q,\omega)\rangle + V_{\perp}\langle \bar{\rho}_\uparrow(q,\omega)\rangle\Big)$$

Assume unpolarized system $\chi_{0\uparrow} = \chi_{0\downarrow} \Rightarrow$

$$\begin{pmatrix} \langle \bar{\rho}_\uparrow(q,\omega)\rangle \\ \langle \bar{\rho}_\downarrow(q,\omega)\rangle \end{pmatrix} = \begin{pmatrix} \chi_0(\bar{q},\omega) & 0 \\ 0 & \chi_0(\bar{q},\omega) \end{pmatrix} \begin{pmatrix} \varphi_\uparrow(q,\omega) \\ \varphi_\downarrow(q,\omega) \end{pmatrix} +$$

$$+ \chi_0(q,\omega) \begin{pmatrix} V_{\parallel}(q) & V_{\perp}(q) \\ V_{\perp}(q) & V_{\parallel}(q) \end{pmatrix} \begin{pmatrix} \langle \bar{\rho}_\uparrow(q,\omega)\rangle \\ \langle \bar{\rho}_\downarrow(q,\omega)\rangle \end{pmatrix}$$

$$\begin{pmatrix} 1 - V_{\parallel}(q)\chi_0(q,\omega) & -V_{\perp}(q)\chi_0(q,\omega) \\ -V_{\perp}(q)\chi_0(q,\omega) & 1 - V_{\parallel}(q)\chi_0(q,\omega) \end{pmatrix} \begin{pmatrix} \langle \bar{\rho}_\uparrow(q,\omega)\rangle \\ \langle \bar{\rho}_\downarrow(q,\omega)\rangle \end{pmatrix} = \chi_0(q,\omega) \begin{pmatrix} \varphi_\uparrow(q,\omega) \\ \varphi_\downarrow(q,\omega) \end{pmatrix}$$

Invert the matrix:

$$\Rightarrow \begin{pmatrix} \langle \rho_\uparrow(q,\omega)\rangle \\ \langle \rho_\downarrow(q,\omega)\rangle \end{pmatrix} = \frac{\chi_0(q,\omega)}{\big(1 - V_{\parallel}(q)\chi_0(q,\omega)\big)^2 - \big(V_{\perp}(q)\chi_0(q,\omega)\big)^2} \cdot$$

$$\cdot \begin{pmatrix} 1 - V_{\parallel}(q)\chi_0(q,\omega) & V_{\perp}(q)\chi_0(q,\omega) \\ V_{\perp}(q)\chi_0(q,\omega) & 1 - V_{\parallel}(q)\chi_0(q,\omega) \end{pmatrix} \cdot \begin{pmatrix} \varphi_\uparrow(q,\omega) \\ \varphi_\downarrow(q,\omega) \end{pmatrix}$$

Density response: $\varphi_\uparrow = \varphi_\downarrow = \varphi(q,\omega)$

$$\begin{pmatrix} \rho_\uparrow \\ \rho_\downarrow \end{pmatrix} = \frac{\chi_0}{(1-V_\parallel \chi_0)^2 - (V_\perp \chi_0)^2} \cdot \begin{pmatrix} 1 - V_\parallel \chi_0 + V_\perp \chi_0 \\ 1 - V_\parallel \chi_0 + V_\perp \chi_0 \end{pmatrix} \cdot \varphi(q,\omega)$$

$\Rightarrow \langle \rho_\uparrow(q,\omega)\rangle = \langle \rho_\downarrow(q,\omega)\rangle$, so that $\langle \rho_\uparrow\rangle - \langle \rho_\downarrow\rangle = 0$,

$$\Rightarrow \langle \rho(q,\omega)\rangle = \langle \rho_\uparrow(q,\omega)\rangle + \langle \rho_\downarrow(q,\omega)\rangle =$$

$$= \frac{2\chi_0(q,\omega)\left[1 - (V_\parallel(q) - V_\perp(q))\chi_0\right]\varphi(q,\omega)}{\left[1 - V_\parallel(q)\chi_0 - V_\perp(q)\chi_0\right]\left[1 - V_\parallel(q)\chi_0 + V_\perp(q)\chi_0\right]} =$$

$$= \frac{2\chi_0(\bar{q},\omega)}{1 - [V_\parallel(q) + V_\perp(q)]\chi_0(\bar{q},\omega)} \varphi(\bar{q},\omega)$$

$$\Rightarrow \chi_\rho(\bar{q},\omega) = \frac{2\chi_0(\bar{q},\omega)}{1 - [V_\parallel(q) + V_\perp(q)]\chi_0(q,\omega)}$$

Coulomb interaction: $V_\parallel(q) = V_\perp(q) = \frac{4\pi e^2}{q^2}$ $\Rightarrow \chi_\rho = \dfrac{2\chi_0}{1 - \frac{4\pi e^2}{q^2}\cdot 2\chi_0}$

Contact interaction: $V_\parallel = 0$, $V_\perp = U$

$$\Rightarrow \chi_\rho = \frac{2\chi_0}{1 - U\chi_0}$$

Spin response : $\varphi_\uparrow = -\varphi_\downarrow = h(\bar{q},\omega)$

$$\begin{pmatrix} \langle \rho_\uparrow \rangle \\ \langle \rho_\downarrow \rangle \end{pmatrix}_{q,\omega} = \frac{\chi_0}{\left[1-(V_\parallel+V_\perp)\chi_0\right]\left[1-(V_\parallel-V_\perp)\chi_0\right]} \cdot$$

$$\cdot \begin{pmatrix} 1-V_\parallel\chi_0 & V_\perp\chi_0 \\ V_\perp\chi_0 & 1-V_\parallel\chi_0 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \end{pmatrix} h(\bar{q},\omega) =$$

$$= \frac{\chi_0}{\left[1-(V_\parallel+V_\perp)\chi_0\right]\left[1-(V_\parallel-V_\perp)\chi_0\right]} \begin{pmatrix} 1-(V_\parallel+V_\perp)\chi_0 \\ -\left(1-(V_\parallel+V_\perp)\chi_0\right) \end{pmatrix} h(\bar{q},\omega)$$

$\Rightarrow \quad \langle \rho_\uparrow(q,\omega)\rangle = -\langle \rho_\downarrow(q,\omega)\rangle$ so that

$\langle \rho_\uparrow \rangle + \langle \rho_\downarrow \rangle = 0 \Longrightarrow$ no charge response

$$\frac{1}{2}\left[\langle \rho_\uparrow(q,\omega)\rangle - \langle \rho_\downarrow(q,\omega)\rangle\right] = \frac{\frac{1}{2}\cdot 2\,\chi_0(\bar{q},\omega)}{1-\left(V_\parallel(q)-V_\perp(q)\right)\chi_0(q,\omega)}\, h(\bar{q},\omega)$$

$$\Rightarrow \quad \chi_\sigma(\bar{q},\omega) = \frac{\chi_0(\bar{q},\omega)}{1-\left[V_\parallel(q)-V_\perp(q)\right]\chi_0(\bar{q},\omega)}$$

Spin susceptibility χ_σ is different from charge χ_ρ one!

Coulomb interaction: $V_\parallel = V_\perp = \frac{4\pi e^2}{q^2} \Rightarrow \chi_\sigma = \chi_0$

Contact interaction: $V_\parallel = 0,\ V_\perp = U \Rightarrow \chi_\sigma = \dfrac{\chi_0}{1+U\chi_0}$